

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457014.87775 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1067 +/- 0.0066
Transit depth (Rp/Rs)^2	1.14 +/- 0.14 [%]
Orbital Inclination (inc)	82.38 +/- 1.97
Airmass coefficient 1 (a1)	1.1625 +/- 0.0067
Airmass coefficient 2 (a2)	-0.1175 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	0.58 %
Best Comparison Star	None
Optimal Aperture	3.65
Optimal Annulus	5.67
Transit Duration (day)	0.1207 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1067 ± 0.0066 vs 0.1178 (deviation 1.43σ)
Duration fit vs published	0.1207 d vs 0.125 d (deviation 0.81σ)
Mid-time O–C	0.2 min
Depth SNR	13.7
Residual scatter	0.58 %

Light curve

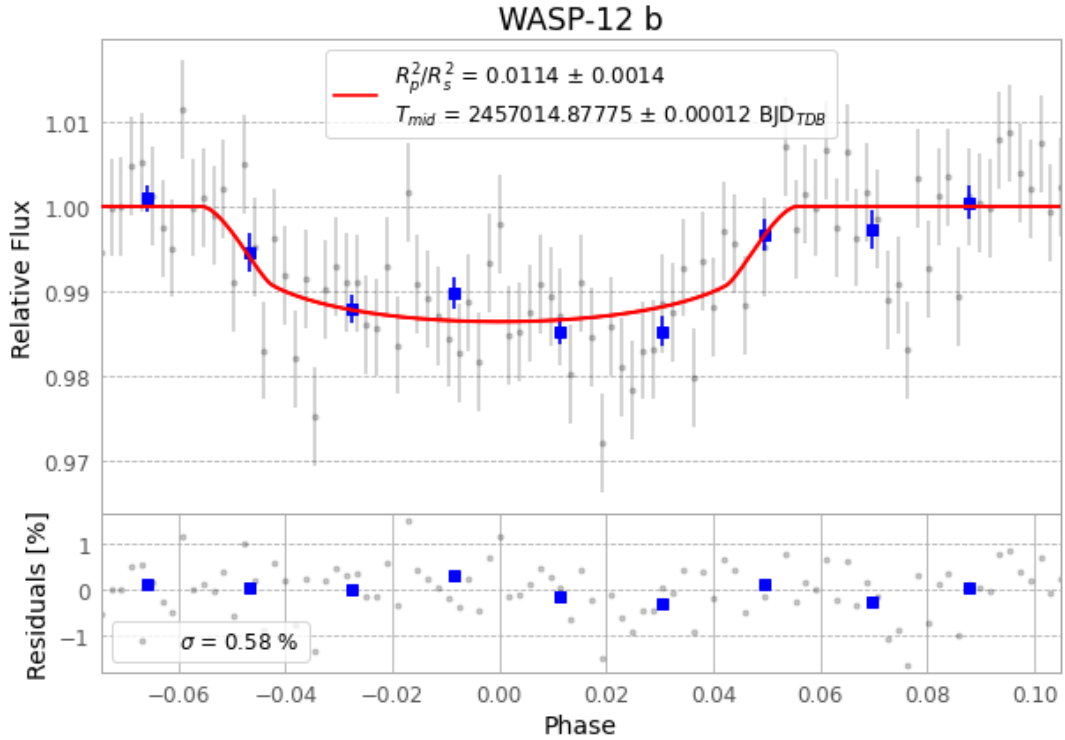


Figure 1 — FinalLightCurve_WASP-12 b_2014-12-22.png (SHA-256 ee22d60a3574b578...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [419, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 850838d365c09a46...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

95 FITS frames (62 MB), WASP-12141223065712.FITS → WASP-12141223113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-141223.log (c17250441199...), FinalLightCurve_WASP-12 b_2014-12-22.png (ee22d60a3574...), FinalLightCurve_WASP-12 b_2014-12-22.csv (c68ecc9f62fe...)

Compute resources

Wall time 123.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)